

Experiment:20

Construct a C program to simulate Reader-Writer problem using semaphores

Code:-

```
#include <stdio.h>

#include <pthread.h>

#include <semaphore.h>

sem_t mutex, writeBlock;

int data = 0; int readersCount = 0;

void *reader(void *arg)

{

for (int i = 0; i < 10; i++) { sem_wait(&mutex); readersCount++;

    if (readersCount == 1)
        sem_wait(&writeBlock);

    sem_post(&mutex);.
    printf("Reader reads data: %d\n", data);

    sem_wait(&mutex);
    readersCount--;

    if (readersCount == 0)
        sem_post(&writeBlock);

    sem_post(&mutex);
}

return NULL;

}

void *writer(void *arg)
```

```
{  
  
    for (int i = 0; i < 10; i++) { sem_wait(&writeBlock);  
  
        data++;  
        printf("Writer writes data: %d\n", data);  
  
        sem_post(&writeBlock);  
    }  
  
    return NULL;  
  
}  
  
int main() { pthread_t r1, r2, w1;  
  
    sem_init(&mutex, 0, 1);  
    sem_init(&writeBlock, 0, 1);  
  
    pthread_create(&r1, NULL, reader, NULL);  
    pthread_create(&r2, NULL, reader, NULL);  
    pthread_create(&w1, NULL, writer, NULL);  
  
    pthread_join(r1, NULL);  
    pthread_join(r2, NULL);  
    pthread_join(w1, NULL);  
  
    sem_destroy(&mutex);  
    sem_destroy(&writeBlock);  
  
    return 0;  
  
}
```

Output:

```
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Writer writes data: 1
Writer writes data: 2
Writer writes data: 3
Writer writes data: 4
Writer writes data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
Writer writes data: 10
Reader reads data: 10
Reader reads data: 10
Reader reads data: 10
Reader reads data: 10
Reader reads data: 10
...Program finished with exit code 0
Press ENTER to exit console.[]
```